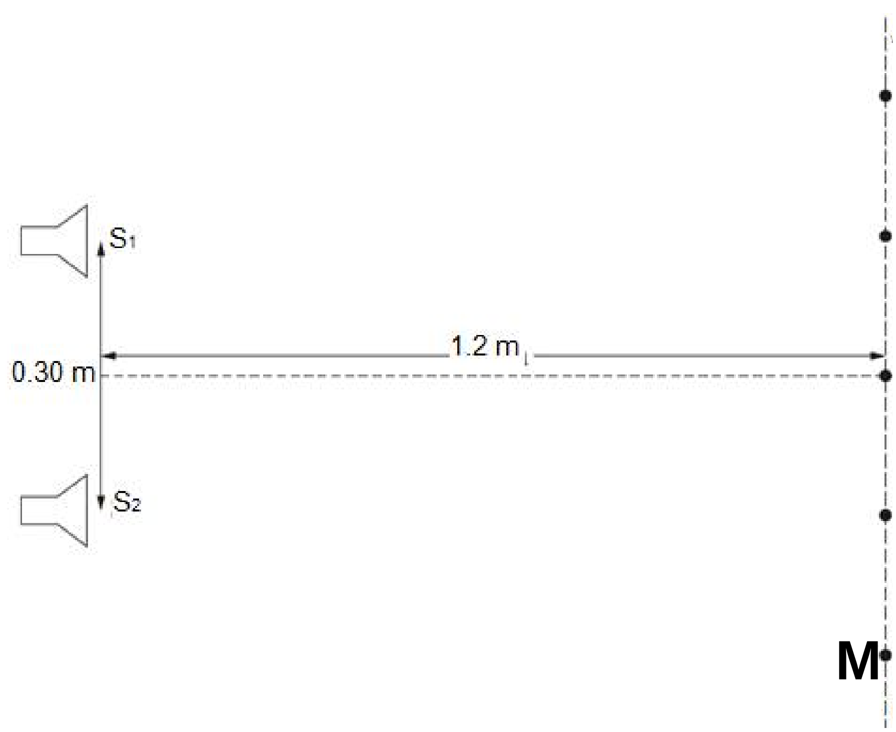
Section B

8	**(a)**	Calculate speed for d = 10, 20, 30 m: Velocity $v_1 = 10 \times \mathbf{2} / 60 \times 10^{-3} = 333.33333$ m s^{-1} $v_2 = 20 \times \mathbf{2} / 120 \times 10^{-3} = 333.33333$ m s^{-1} $v_3 = 30 \times \mathbf{2} / 170 \times 10^{-3} = 352.94$ m s^{-1} Explanation: The value of d is multiplied by 2 (or half the t) because the sound travels from the source to wall and back again to source. Find the average. : $v_{ave} = 339.87 = 340$ m s^{-1} [Alternatively, students can solve via drawing a BFL, determining its gradient, equating the gradient value to 2/v and then find v. Drawing a BFL is akin to taking average.]	 **1** **1** **1**
	(b)	**(i)** Use $\lambda = ax/D$ With x = 0.16 m, …. λ = 0.040 m	 **1** **1**
	(b)	**(ii)** $v = f\lambda = 8300 \times 0.040$ = 332 m s^{-1}	 **1**
		(iii) **1.** 	**1**

			2. The path difference is equal to an <u>integer</u> number of wavelengths (and sources are in phase) OR Waves from S_1 and S_2 <u>arrive in phase at **M**</u> ["Constructive interefence" as the sole answer will not gain any credit. This is just paraphrasing the question]	**1**
		(iv)	Calculate that λ = 1.1 m Use $\lambda = ax/D$ to calculate x = 4.4 m The first order maximum is found at 4.4 m which is too far away from the central fringe. (The length of the screen AB appears to be slightly greater than 0.64 m only. So there is a central maxima, but the next maxima is 4.4 m away, and the first minima is 2.2 m away, so no succession of maxima and minima).	**1** **1** **1**
	(c)	**(i)**	$d = 1.0 \times 10^{-4}$ m $n\lambda = d\sin\theta$ With n = 1, θ = 0.286°	**1** **1**
		(ii)	For λ = 700 nm, n = 1, we have θ_{700} = 0.4011° Thus $\Delta\theta = \theta_{700} - \theta_{500}$ = 0.4011 – 0.2865 = 0.115°	**1** **1**
		(iii)	Let y be the distance of the maxima from **X**, as measured along the screen. Then y_{700} = 2.0 tan 0.4011° and y_{500} = 2.0 tan 0.2865°. Thus, linear separation = $y_{700} - y_{500}$ = 0.00400 m	 **1** **1**
		(iv)	lower intensity (because energy spreads) use or statement of inverse square law ($I \propto 1/r^2$) ratio of the initial intensity to the final intensity = $(0.02/0.05)^2$ = 0.16 or intensity falls by factor of 6.25	**1** **1** **1**

9	(a)	(i)		